Belief Revision in a Mixed Fleet

A Ranking per Robot, Belief Reports Weighed by Trust,
and a Courier that Chooses What to Tell, on ePlanSys

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Robots of different makers that share a floor revise their beliefs by different orders: the same report of evidence leaves them believing different things, and each can know what the others believe. This report carries a Spohn ranking per robot in the epistemic state an executor maintains, as grade atoms in every world, and adds the one action that changes a ranking: a report of belief, after which a listener that takes the speaker's word raises its rank of every world the report rules out. We prove that belief, nested belief included, is a formula over knowledge and each robot's own grades; that evidence revises each robot by Spohn conditioning on its own ranking; that a belief report is Spohn conditioning on evidence of a given strength and is written exactly by one assignment per grade atom; that robots sharing a ranking cannot commonly know they disagree, while robots that do not share one can; and that a courier choosing its reports by a rollout of the policy without reports never raises the fleet's predicted expected time. A courier that takes the others to rank as it does has nothing to say. On a hospital floor in Gazebo, a night porter shuts the two fire doors of a loop no robot sees him walk, and a fleet of the hospital's courier, a disinfection robot that ranks door by door and a linen robot that ranks nothing crosses the floor in three acts, the courier silent, telling every belief its listener lacks, or telling what it predicts will help. Every robot reached its bed in all 108 runs. Telling what it predicted would help, the courier brought the fleet in sooner in expectation in acts I and III, 265.8 and 304.6 s against 282.5 and 384.2 s silent; on the west-loop night of act I, 309 s against 463. In act II its one report at dusk was predicted to help and did not: 310.2 s against 306.4, because ePlanSys schedules the two robots more serially than the planner assumed. Telling every belief the listener lacks gained in act I and nothing in acts II and III (301.9 and 386.9 s).

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robot	maker	ranking of the loops	route	takes the word of
c	the hospital	the porter's habits: CD 0, AB 1, others 2	day room to ward 4	nobody
d	a disinfection contractor	door by door: CD 0, AB 4, others 2	pharmacy to ward 2	*c* (acts I, III)
e	a linen contractor	none: every loop 0	treatment room to ward 1	*c* (acts II, III)

Table 1: The fleet. Door by door ranks a loop by the number of doors whose state differs from the rostered picture, Dalal's distance [6], which is how a map of independent cells ranks a change. The linen robot believes only what it knows. Every robot knows every robot's ranking and who takes whose word.

1 Scope

The companion report on belief revision in a changing ward [1] placed a Spohn ranking in ePlanSys's epistemic state and showed that every observation then revises by conditioning. Its couriers shared one ranking, received only hard information, and planned each for itself; its limitations named all three. A fleet in a hospital rarely comes from one maker. The robot that carries drugs is the hospital's; the one that disinfects belongs to a contractor whose software keeps a map of independent cells; the one that carries linen belongs to another and keeps no prior at all. They meet on the same floor, report what they find on a shared channel, and revise on the reports, each in its own way.

Three questions follow, and the report answers them in this order. What does the epistemic state need so that each robot revises by its own ranking, and how do nested beliefs read off it? What is a report of belief, which a robot may make only sincerely and which a listener may weigh by how far it trusts the speaker, as an event the executor can apply? And when should a robot that knows how the others revise tell them what it believes? The first two are answered by propositions about ranked models and their compilation into the EPDDL that plank grounds; the third by a planner and a batch of 108 runs in Gazebo, compared under three formulations of what the courier says.

Nothing in plank [21], Aletheia or ePlanSys was changed. The runs use the executor patch distributed with the companion demonstration.

2 The ward and the fleet

The floor is the companion report's ward: a single storey of 28 by 16 m with rooms on the south side of a hall, a band of linen rooms crossed by four fire lobbies, A to D from west to east, a corridor north of the band and four ward bays off it. Each lobby turns once, and its door stands across the jog, so a door is decided only by a laser at a bend inside its lobby; the floor plan is checked for this in all sixteen states of the doors, and every route keeps 0.47 m from anything a robot can hit.

At one in the morning the night porter walks one loop through the band, north through one door and south through another, and shuts both. No robot sees him. On most nights he walks the rostered loop through C and D, on some the west loop through A and B, and on a few one of the other four; the expectations in this report weigh the nights by those frequencies, 0.842, 0.105 and 0.013 each.

3 Ranked models with many robots

3.1 Models, events and updates

Definition 1 (Epistemic model [12])**.** An epistemic model over atoms P and agents $\mathcal{A}$ is $\mathcal{M} = (W, \{\sim_r\}_{r\in\mathcal{A}}, V)$ with W finite, each $\sim_r$ an equivalence and $V : W \to 2^P$. A state is a model with designated worlds $W_d \subseteq W$. We write $[w]_r$ for r's class of w and K_r, $\hat{K}_r$ for its box and diamond.

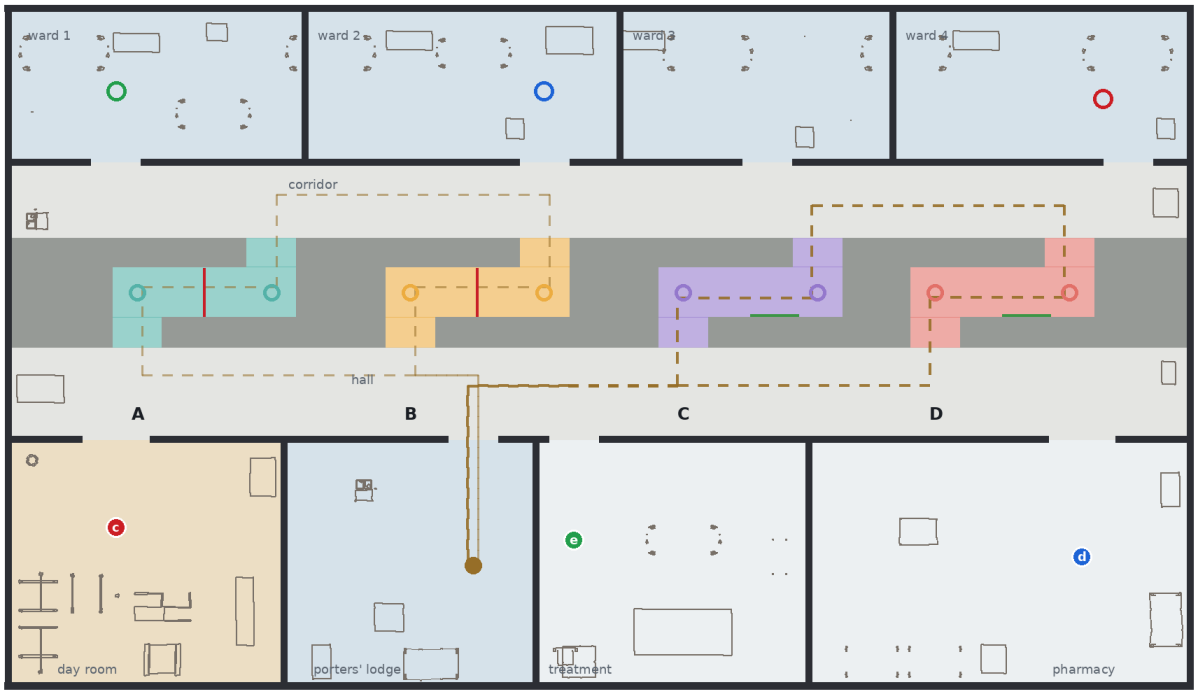


Figure 1: The ward from above. The lobbies are tinted, A teal, B amber, C violet and D coral, and a ring marks the bend from which each door is seen. The robots start in the day room (the courier, red), the pharmacy (the disinfection robot, blue) and the treatment room (the linen robot, green), and deliver to the beds marked in their colours, in wards 4, 2 and 1.

Definition 2 (Event model and product update [11, 12])**.** An event model is $\mathcal{E} = (E, \{\sim_r^E\}, pre, post)$ with $pre(e)$ a formula and $post(e)$ assigning a formula to some atoms. The product $\mathcal{M} \otimes \mathcal{E}$ has worlds $\{(w,e) : \mathcal{M}, w \models pre(e)\}$, relations $(w,e) \sim_r (v,f)$ iff $w \sim_r v$ and $e \sim_r^E f$, and $p \in V(w,e)$ iff $\mathcal{M}, w \models post(e)(p)$ where $post(e)$ assigns p, and iff $p \in V(w)$ otherwise.

3.2 A ranking per robot

Definition 3 (Multi-agent ranked model)**.** A ranked model is an epistemic model with a ranking $\kappa_r : W \to \mathbb{N}$ for every agent r. For $X \subseteq W$ let $\min_{\kappa_r} X$ be the worlds of X of least κ_r. Then $\mathcal{M}, w \models B_r\varphi$ iff $\mathcal{M}, v \models \varphi$ for every $v \in \min_{\kappa_r}[w]_r$.

This is a plausibility model in the sense of Baltag and Smets [7] with one order per agent, total within each of its cells and given by numbers as in Spohn's ordinal conditional functions [2]. The formula φ may itself contain other agents' belief operators: $B_c\neg B_d\varphi$ is "c believes that d does not believe φ".

Definition 4 (Grade compilation)**.** Fix N and fresh atoms $g_0^r, \dots, g_N^r$ for every agent r. The compilation $\mathcal{M}^g$ has $V^g(w) = V(w) \cup \{g^r_{\kappa_r(w)} : r \in \mathcal{A}\}$, provided every $\kappa_r \le N$. Define the translation β on formulas, commuting with the Boolean connectives and the K_s, by

$$\beta(B_r\varphi) = \bigvee_{k=0}^{N} \Big(K_r \textstyle\bigwedge_{j<k} \neg g_j^r \wedge \hat{K}_r\, g_k^r \wedge K_r\big(g_k^r \to \beta(\varphi)\big)\Big).$$

Proposition 1 (Belief as a formula, for every robot)**.** For every world w and every formula φ of the language with all agents' belief and knowledge operators, $\mathcal{M}, w \models \varphi$ iff $\mathcal{M}^g, w \models \beta(\varphi)$. Each B_r satisfies KD45, each K_r S5, $K_r\varphi \to B_r\varphi$, and $B_r\varphi \to K_rB_r\varphi$.

Proof. By induction on φ. The Boolean and knowledge cases are immediate, since compilation changes no world or relation. For $B_r\psi$, let $C = [w]_r$ and $m = \min_{v\in C} \kappa_r(v)$. For $k < m$ the

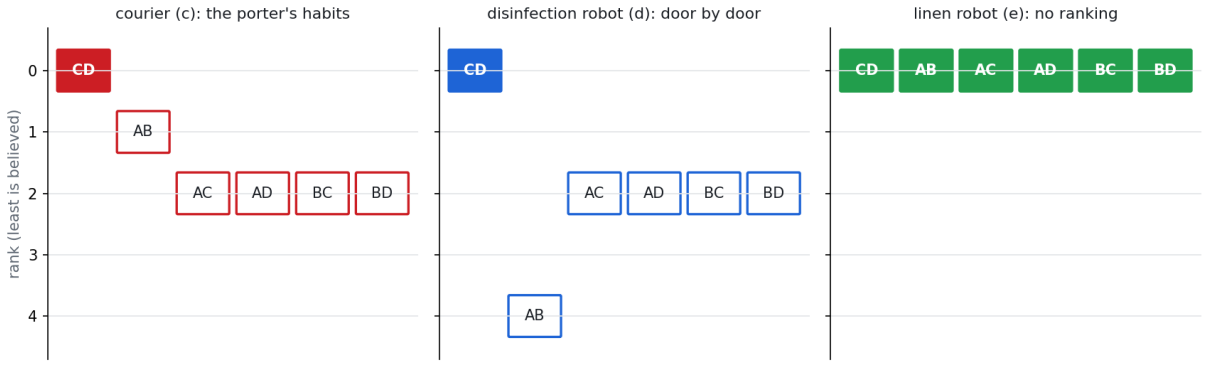


Figure 2: The six loops as each robot ranks them at dusk, the believed loops filled. The courier and the disinfection robot both believe the rostered loop and differ in what each will believe once it is ruled out; the linen robot believes nothing beyond the six.

conjunct $\hat{K}_r g^r_k$ fails, for $k > m$ the conjunct $K_r \bigwedge_{j<k} \neg g^r_j$ fails, and for $k = m$ the first two hold and the third says that $\beta(\psi)$, hence by induction ψ, holds at every world of C of grade m, which is $\min_{\kappa_r} C$. The grades of r are atoms, the same at a world whoever considers it, so another agent's operator applied to $B_r\psi$ evaluates r's belief at each world of its own cell on r's cell there. $\min_{\kappa_r} C$ is non-empty, which gives D; it depends on C alone, which gives 4, 5 and the last claim; $\min_{\kappa_r} C \subseteq C$ gives the third. □

The domain needs β of depth two: the guard of a belief report says that the speaker believes the listener does not believe what it reports (Section 5). With $N = 9$ the executor evaluates that formula on six worlds.

4 Evidence

4.1 Each robot conditions on its own ranking

Proposition 2 (Evidence revises each robot by its own ranking)**.** Let $\mathcal{E}$ assign no grade atom, and set $\kappa_r(w, e) = \kappa_r(w)$. Then $(\mathcal{M} \otimes \mathcal{E})^g = \mathcal{M}^g \otimes \mathcal{E}$, and at each world of the update every robot r believes on $\min_{\kappa_r}$ of its new cell, which is Spohn's conditioning of κ_r on that cell. For each robot, a public event with a precondition consistent with its knowledge defines an AGM revision of its belief set [4], by the faithful order κ_r [5].

Proof. As in the companion report [1], for each r separately: the product keeps every grade atom, so each world of the update carries the grades of the world it came from, and Proposition 1 reads r's belief on the least grade of its new cell. The order $v \leq_r u$ iff $\kappa_r(v) \leq \kappa_r(u)$ is a total preorder on r's cell faithful to r's belief set, and Katsuno and Mendelzon's representation gives the AGM postulates. □

Looks, reports of evidence and crossings are such events. A report of a door found shut is a public announcement of $\neg open_x \wedge K_r \neg open_x$; every robot conditions on it, and the robots end with the same cell and different beliefs. On the west-loop night of act I the courier finds B shut and reports it. It then believes the west loop, and so A shut as well, while the disinfection robot, whose least loops left are B-C and B-D, believes A open.

4.2 Agreement, and when it fails

Aumann showed that agents with a common prior cannot agree to disagree: if their posteriors for an event are common knowledge, they are equal [13]. Bacharach derived it for any decision function that is the same for all agents and satisfies a sure-thing principle [14]; Dégremont and Roy gave versions in dynamic epistemic logic [15]. Belief read off a ranking is such a function.

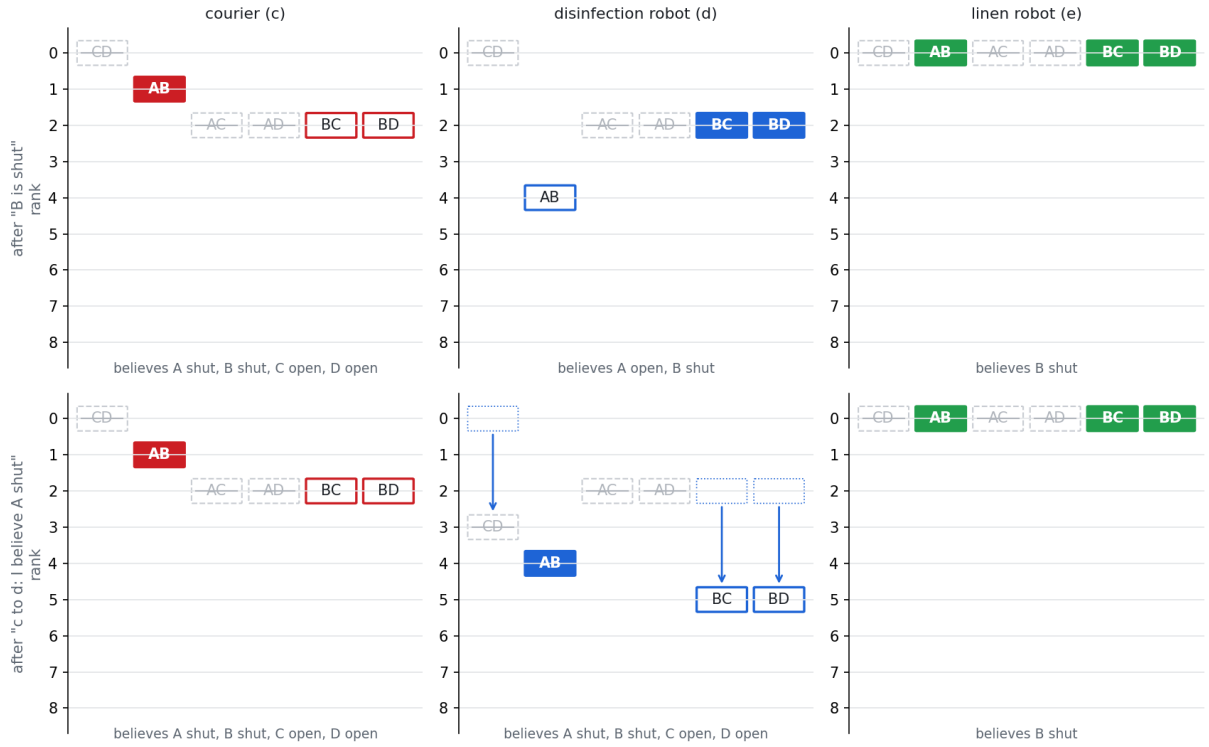


Figure 3: Act III, the west-loop night, the courier silent at dusk. Top: after the courier's report of B shut, ruled-out loops struck through and believed loops filled; three robots, three beliefs about A. Bottom: after the courier tells the disinfection robot that it believes A shut; that robot's ranks of B-C and B-D rise by $\alpha = 3$ (the arrows) and its belief becomes the courier's. The linen robot was not told.

Proposition 3 (Agreement under a common ranking)**.** Let every robot of a group G rank by the same κ, and let $D(X) \in \{\text{yes}, \text{no}, \text{neither}\}$ say whether $\min_\kappa X$ satisfies φ, satisfies $\neg\varphi$, or neither. If at w it is common knowledge in G that each robot $r \in G$ has $D([\cdot]_r) = v_r$, then all the v_r are equal.

Proof. D is union consistent: if $X \cap Y = \emptyset$ and $D(X) = D(Y) = v$, then $D(X \cup Y) = v$. For if $\min_X \kappa < \min_Y \kappa$ then $\min_\kappa(X \cup Y) = \min_\kappa X$; and if the minima are equal then $\min_\kappa(X \cup Y) = \min_\kappa X \cup \min_\kappa Y$, on which φ holds throughout, fails throughout, or does both somewhere, exactly as it does on each part. Let M be the component of w under the union of the relations of G. For each r, M is a disjoint union of r's cells, on each of which D is v_r by the common knowledge, so by union consistency $D(M) = v_r$. Hence $v_r = D(M)$ for every r. □

Fact 1 (Commonly known disagreement)**.** In act III, with the silent courier, after its report of B shut on the west-loop night, the state the product update of the grounded task produces satisfies $C_{\{c,d,e\}}\big(B_c\neg open_a \wedge B_d open_a \wedge \neg B_e open_a \wedge \neg B_e\neg open_a\big)$: every robot knows what each believes about door A, and they believe three different things. The same holds in act I for the courier and the disinfection robot. Checked by `tools/agreement.py`, which reads each robot's belief off its grade atoms at every world reachable from the designated ones.

The premise that fails is the common ranking. The companion report's Proposition 5, that a report of evidence carries the belief, assumed it; in a mixed fleet a report of evidence carries the evidence and nothing more.

5 Testimony

5.1 A report of belief

A robot s may report that it believes φ, for φ an atom or the negation of one. A listener l that takes s's word with trust α raises its rank of every world where φ is false by α, as if the report were evidence that would be wrong only with rank α; a robot that does not take s's word is not moved. Everyone hears the report.

Definition 5 (Belief report)**.** $\mathsf{believe}_{s\to l}(\varphi)$ is the event model with one public event of precondition

$$\pi_{s,l}(\varphi) = \mathit{takes}(l,s) \wedge B_s\varphi \wedge \neg K_s\varphi \wedge B_s\neg B_l\varphi$$

and postcondition, for every grade $t < N$ and for N,

$$g^l_t := (\varphi \wedge g^l_t) \vee \big(\neg\varphi \wedge g^l_{t-\alpha}\big), \qquad g^l_N := g^l_N \vee \Big(\neg\varphi \wedge \textstyle\bigvee_{k\geq N-\alpha} g^l_k\Big),$$

with $g^l_{t-\alpha}$ read as false when $t < \alpha$.

The precondition is sincerity (the speaker believes it), news to its own knowledge (else it reports the evidence), and relevance in Grice's sense [16]: the speaker believes that the listener does not already believe it. The last conjunct is a second-order belief, and the executor checks it before it dispatches the report.

Proposition 4 (A belief report is conditioning on evidence of strength α)**.** Let $\kappa_l + \alpha \leq N$ on the worlds where φ fails. The update of $\mathcal{M}^g$ by $\mathsf{believe}_{s\to l}(\varphi)$ keeps the worlds where $\pi_{s,l}(\varphi)$ holds, changes no relation, no other robot's grade and no other atom, and gives l the ranking

$$\kappa'_l(w) = \kappa_l(w) + \alpha\,[\,w \not\models \varphi\,].$$

Moreover:

1. this is the product of $\mathcal{M}$ with the two-event model $\{e_1, e_0\}$, $\mathit{pre}(e_1) = \varphi$, $\mathit{pre}(e_0) = \neg\varphi$, confused by every agent and ranked 0 and α for l, with ranks added [9, 10, 7]; and it is Spohn's conditioning on evidence of strength α in Shenoy's form [3, 2];
2. two reports commute, and k reports of φ are one of strength $k\alpha$;
3. after the report l believes φ at w iff $\min_{C\cap\neg\varphi}\kappa_l + \alpha > \min_{C\cap\varphi}\kappa_l$, $C = [w]_l$;
4. if l later learns φ, or learns $\neg\varphi$, its beliefs are those it would hold had the report not been made.

Proof. The event is public with one event, so the product keeps exactly the worlds satisfying its precondition and every relation between them. At a world where φ holds every assignment returns g^l_t, so no grade moves. At a world where φ fails and l's grade is k, exactly one grade atom of l is true, g^l_k, and the assignment for t is true iff $t = k + \alpha$. (1) The product with $\{e_1, e_0\}$ has one copy of each world, (w, e_1) if φ holds and (w, e_0) otherwise, related as the worlds were, and the added ranks are $\kappa_l(w) + \alpha[w \not\models \varphi]$. Shenoy's conditioning on φ with strength α adds α to the rank of the worlds outside φ and renormalises, and belief depends only on the least worlds, which renormalisation keeps. (2) Addition commutes. (3) $\min'_{\kappa_l} C$ lies in φ iff the least rank on $C \cap \varphi$, unchanged, is below the least rank on $C \cap \neg\varphi$, raised by α. (4) Learning φ restricts C to φ-worlds, on which $\kappa'_l = \kappa_l$; learning $\neg\varphi$ restricts it to $\neg\varphi$-worlds, on which $\kappa'_l = \kappa_l + \alpha$, a constant shift that leaves the least worlds where they were. □

Commutation distinguishes the graded report from the radical upgrade of van Benthem and of Baltag and Smets [8, 7], after which the last report wins. A listener here weighs each report by its trust and adds them; a report and its denial, from speakers it trusts alike, cancel.

Proposition 5 (What hearing a report teaches)**.** Hearing $\mathsf{believe}_{s\to l}(\varphi)$ removes from every robot's cell exactly the worlds where s would not have made it. If s's cell at the designated worlds is every other robot's cell there, it removes none of them.

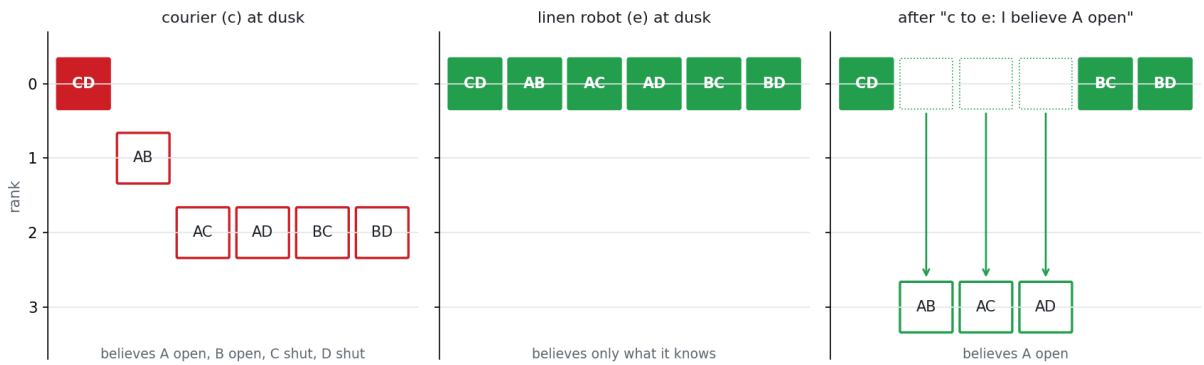


Figure 4: Act II at dusk. The linen robot ranks nothing; told "I believe door A is open", it raises the three loops through A by three and believes A open, and nothing yet about the other doors.

Proof. The first claim is the product with a single public event. B_s, K_s and $B_s\neg B_l$ are constant on s's cells (Proposition 1), and *takes* is the same at every world, so $\pi_{s,l}(\varphi)$ is constant on s's cell; if that cell is every robot's cell, the precondition is true throughout it, since it is true at the designated worlds where the report is made. □

In the runs every robot reports at once a door it finds shut and crosses a door it finds open in everyone's sight, so the courier's evidence is public whenever it speaks, and a belief report teaches nothing about the floor: it moves the listener through its trust alone. Of the sixteen belief reports in the relevance and perspective policies, none removes a world. A robot that knew the courier's ranking and chose to defer to it could reach the same belief unprompted. The contractors' robots do not: they revise on what they are told.

5.2 The encoding in EPDDL

The postcondition of a belief report assigns every grade atom of the listener. A first version wrote it as conditional effects, one per grade k, moving a world of grade k where φ fails to grade $k+\alpha$: `(when (and` $\neg\varphi$ `(rank`k `?l)) (and (not (rank`k `?l)) (rank`$k{+}\alpha$ `?l)))`. plank compiles several conditional effects on one atom into the conjunction of their assignments, $p := (p \vee c_1) \wedge (p \wedge \neg c_2)$ for an atom one adds and another deletes, which loses the addition: after the report the listener had no grade at the shifted worlds. `tools/policy_check.py` found it at the first belief report it replayed. The domain now writes one `iff` per grade atom, an assignment plank compiles as written; for the report that door A is shut, grade 5 and $\alpha = 3$:

```
(iff (or (and (not (open-a)) (rank5 ?l)) (and (open-a) (rank2 ?l))) (rank5 ?l))
```

6 The domain

The EPDDL domain is written by `tools/model.py`. Its atoms are the doors' states, the grades $rank_k(r)$ for $k \leq 9$, each robot's place, whether it is still searching, whether it has delivered, and $takes(l, s)$. Its actions, for a robot r:

- set-out from its room to the hall;
- seek, a hop along the hall, guarded by $\neg B_r$ (every door ahead shut);
- enter the lobby of x to the bend, guarded by $\neg B_r \neg open_x$, and back out to the mouth;
- look at x: semi-private sensing, r fully, the others partially;
- cross through x, requiring $open_x \wedge K_r\, open_x$, after which r is no longer searching, and deliver along the corridor to its bed;
- tell x, a report of evidence: public, with precondition $\neg open_x \wedge K_r \neg open_x$;
- believe x shut, or open, from s to l: the belief report of Section 5.

Each act's problem gives every robot the universal relation over the six loops, labels each loop with every robot's grade (Table 1) and states who takes whose word. Grounded by plank, the three problems have 100, 100, 174 actions. The executor's vocabulary maps them to five classical actions, drive, lobby, look, tell and believe, whose facts exist only for ePlanSys's parallel groups: a lobby action holds its lobby, so two robots are never dispatched into one together, and a report writes a fact every action reads, so it goes alone.

7 Planning

7.1 Each robot routes itself

A robot chooses its next move from what it knows and how it ranks the worlds it cannot rule out. Its objective is the time it expects to take to its bed, each world of its cell weighted $\varepsilon^{\kappa_r(w)}$ with $\varepsilon = 1/8$, the reading of ranks as orders of magnitude [17]; its own looks branch the plan, and it never drives to a door it believes shut. The linen robot, every loop of rank 0, counts the loops alike. A dynamic programme over the robot's place and cell finds the optimum; the robot does not anticipate what it will be told. Every robot routes so, the courier included, and every robot reports a door it finds shut at once.

Ranks are orders of magnitude here, and not only an order: a report too weak to change what the disinfection robot believes can still change where it goes, because it changes the weights of the loops it does not believe (Section 10).

7.2 What the courier says

At each of its steps the courier may instead make a belief report its guard allows. The three formulations differ only in which:

- *egocentric*: the courier takes every robot to rank as it does, and evaluates the relevance of a report in that projected model;
- *relevance*: it makes every report its guard allows, as soon as the guard allows it, each once to each listener;
- *perspective*: it makes a report when the fleet's expected time to the last delivery is lower with it, each world of its cell weighted $\varepsilon^{\kappa_c(w)}$, the others predicted on the rankings and trust they have. The prediction runs every robot's routing to the end with the courier making no further report: a rollout of the policy without reports, one step of lookahead [18].

Proposition 6 (Egocentric silence)**.** In the projected model, where every robot ranks by κ_c, suppose that for every robot t still searching, every designated w and every $v \in \min_{\kappa_c}[w]_c$, $\min_{\kappa_c}[v]_t = \min_{\kappa_c}[w]_c$. Then the guard $B_c\neg B_t\varphi$ fails for every φ the courier believes, and the egocentric courier makes no report. In particular it makes none when every robot's cell is the courier's.

Proof. At every v the courier believes possible, t believes on $\min_{\kappa_c}[v]_t = \min_{\kappa_c}[w]_c$, the worlds on which the courier believes φ; so $B_t\varphi$ holds at every such v, and $B_c\neg B_t\varphi$ fails. □

`tools/validate.sh` checks that the egocentric policy is, in every act, the policy in which nobody reports a belief.

Proposition 7 (Rollout improvement)**.** Let V_π be the expected predicted time to the last delivery of a policy π, each night weighted by the porter's frequency. Then $V_{\text{perspective}} \le V_{\text{egocentric}}$ in every act.

Proof. The frequencies are proportional to ε^{κ_c}, so at a state where the courier chooses, its weights are the frequencies conditioned on its cell, and the value it assigns to an option is the expected time of taking it and then following the base policy b without reports, which is a function of the state and the actual night. The base action is among the options. By induction on the number

night	egocentric	s	relevance	s	perspective	s
CD	nothing	258	nothing	258	nothing	258
AB	c: B shut; d: A shut	463	c: B shut; c→d: “A shut”	309	c: B shut; c→d: “A shut”	309
AC	nothing	259	nothing	259	nothing	259
AD	nothing	259	nothing	259	nothing	259
BC	c: B shut	387	c: B shut; c→d: “A shut”; d: C shut	359	c: B shut; c→d: “A shut”; d: C shut	359
BD	c: B shut	313	c: B shut; c→d: “A shut”	308	c: B shut; c→d: “A shut”	308

Table 2: Act I: the reports each formulation’s policy makes on each night, as planned, and the measured mission time, the mean of the night’s runs. x: B shut is a report of evidence by robot x; c→d: “A shut” a report of belief to the disinfection robot.

of the courier’s remaining choices, the rollout’s expected time from any such state is at most b’s: at its last choice it picks an option no worse than b’s by construction, and at an earlier one it picks the option whose b-continuation is least, which its own continuation does not exceed by the induction hypothesis. At the root this is the claim, and the egocentric policy is b (Proposition 6 and the check above). □

The bound is on predicted times, the planner’s model of ePlanSys’s parallel groups; whether it holds on the floor is what the batch measures. No such bound holds for relevance, and the predictions show it failing: in acts II and III, telling the linen robot every belief it lacks costs more than telling it nothing (Section 8).

The team’s policy is one tree, built in the order ePlanSys dispatches it: at each step the robots choose in turn, each in the state the earlier ones’ actions left, a look ends its group because it branches, and a report goes alone. `tools/plan.py` writes it in Aletheia’s policy format; the AletheiaPlanSolver plugin calls it as its planner and validates the policy against the grounded task, and `tools/policy_check.py` replays every policy through the repository’s product update and checks, at every node, that every robot’s cell, grades and belief read off the model are those of the reference semantics: 3 888 checks over the twelve policies, with no disagreement and no grade at its cap.

8 The three acts

Act I, the courier and the disinfection robot. The courier leaves the day room for ward 4 and the disinfection robot the pharmacy for ward 2. At dusk both believe the rostered loop, so A and B open, and both make for B, the courier from the west and the disinfection robot from the east. On the west-loop night the courier enters lobby B first, finds the door shut and reports it. It now believes the west loop, so A shut as well; the disinfection robot, whose two least loops left are B-C and B-D, believes A open, and both know it. The egocentric courier, taking the other to rank as it does, says nothing, and the disinfection robot drives on to A, finds it shut, and only then goes east to C: 463.2 s on that night. Under relevance and perspective the courier says "I believe door A is shut". The disinfection robot raises B-C and B-D from rank 2 to 5, now believes the west loop too, and turns east for C: 308.6 s. On the rare nights B-C and B-D the report is wrong, A is open, and the disinfection robot is sent the long way.

Act II, the courier and the linen robot. The courier and the linen robot, which ranks nothing. At dusk the linen robot believes only that one of the six loops was walked; B is its nearer door, and it would make for it and wait in the hall while the courier, which also makes for B, goes through the lobby. Under perspective the courier tells it, before anyone moves, "I believe door A is open": the linen robot raises the three loops through A to rank 3, believes A open, and goes to A while the courier crosses at B. The courier predicted 289 s on the rostered night against 328 s for the silent courier; on the floor it took 296.9 s against 291.2 s. The wait at B it was to spare cost less than the planner’s model of the executor’s groups said, and the drive to A cost about as much

night	egocentric	s	relevance	s	perspective	s
CD	nothing	291	c→e: “A open”; c→e: “B open”	288	c→e: “A open”	297
AB	c: B shut; e: A shut	425	c→e: “A open”; c→e: “B open”; c: B shut; c→e: “A shut”; c→e: “C open”; c→e: “D open”	389	c→e: “A open”; c: B shut; e: A shut	397
AC	nothing	288	c→e: “A open”; c→e: “B open”	294	c→e: “A open”	379
AD	nothing	291	c→e: “A open”; c→e: “B open”	295	c→e: “A open”	382
BC	c: B shut	344	c→e: “A open”; c→e: “B open”; c: B shut; c→e: “A shut”; c→e: “C open”; c→e: “D open”; e: C shut	387	c→e: “A open”; c: B shut	326
BD	c: B shut	321	c→e: “A open”; c→e: “B open”; c: B shut; c→e: “A shut”; c→e: “C open”; c→e: “D open”	388	c→e: “A open”; c: B shut	312

Table 3: Act II: the reports each formulation's policy makes on each night, as planned, and the measured mission time, the mean of the night's runs. x: B shut is a report of evidence by robot x; c→d: “A shut” a report of belief to the disinfection robot.

night	egocentric	s	relevance	s	perspective	s
CD	nothing	360	c→e: “A open”; c→e: “B open”	368	c→e: “A open”	283
AB	c: B shut; e: A shut	569	c→e: “A open”; c→e: “B open”; c: B shut; c→d: “A shut”; c→e: “A shut”; c→e: “C open”; c→e: “D open”	504	c→e: “A open”; c: B shut; c→d: “A shut”; e: A shut	440
AC	nothing	413	c→e: “A open”; c→e: “B open”	363	c→e: “A open”; e: A shut	396
AD	nothing	363	c→e: “A open”; c→e: “B open”	404	c→e: “A open”; e: A shut	395
BC	c: B shut; c: C shut	457	c→e: “A open”; c→e: “B open”; c: B shut; c→d: “A shut”; c→e: “A shut”; c→e: “C open”; c→e: “D open”; d: C shut	548	c→e: “A open”; c: B shut; c→d: “A shut”; d: C shut	372
BD	c: B shut	388	c→e: “A open”; c→e: “B open”; c: B shut; c→d: “A shut”; c→e: “A shut”; c→e: “C open”; c→e: “D open”	516	c→e: “A open”; c: B shut; c→d: “A shut”	343

Table 4: Act III: the reports each formulation's policy makes on each night, as planned, and the measured mission time, the mean of the night's runs. x: B shut is a report of evidence by robot x; c→d: “A shut” a report of belief to the disinfection robot.

(see the evaluation). Under relevance the courier tells it every belief it lacks, A open and B open both: the linen robot believes both, takes the nearer, B, and waits behind the courier, two reports later: 288.5 s. On the west-loop night, and on A-C and A-D, the report "A open" is wrong, and the linen robot finds A shut itself.

Act III, all three. All three robots, both contractors taking the courier's word. On the rostered night the silent courier leaves three robots making for lobby B, each waiting for the one before it; under perspective it tells the linen robot "I believe door A is open", as in act II, and the night takes 283.2 s against 359.8 s. On the west-loop night the courier finds B shut and reports it, and the three robots are left with three beliefs about A, commonly known. The courier tells the disinfection robot it believes A shut, as in act I; it does not tell the linen robot, which it told A open at dusk and which, heading for A, will see for itself: 440.0 s against 568.6 s. Relevance tells the linen robot A open and B open at dusk, and after the report A shut, C open and D open, five belief reports in all: 504.1 s.

9 The batch

In act I the expected times were 282.5 s egocentric, 265.8 s relevance and 265.8 s perspective. In act II the expected times were 306.4 s egocentric, 301.9 s relevance and 310.2 s perspective. In act III the expected times were 384.2 s egocentric, 386.9 s relevance and 304.6 s perspective.

act	formulation	expected	predicted	rostered	west	arrived
I	egocentric	282.5	316.8	258.5	463.2	12/12
	relevance	**265.8**	308.6	258.5	308.6	12/12
	perspective	265.8	308.6	258.5	308.6	12/12
II	egocentric	306.4	342.4	291.2	425.4	12/12
	relevance	**301.9**	344.2	288.5	389.2	12/12
	perspective	310.2	309.0	296.9	396.6	12/12
III	egocentric	384.2	431.0	359.8	568.6	12/12
	relevance	386.9	432.6	367.8	504.1	12/12
	perspective	**304.6**	332.9	283.2	440.0	12/12

Table 5: Expected mission time in seconds, the mean of each night's runs weighted by the porter's frequencies, the runs of formulations that plan the same path on a night pooled, and the same for the times `tools/plan.py` predicts; the means on the rostered and west-loop nights; runs in which every robot reached its bed. In bold, the least expected time of each act.

act	formulation	planner's model	ePlanSys's schedule	measured
I	relevance	+8.2	+12.7	+16.7
	perspective	+8.2	+12.7	+16.7
II	relevance	-1.8	-4.9	+4.5
	perspective	+33.4	+5.1	-3.8
III	relevance	-1.6	+5.0	-2.7
	perspective	+98.1	+72.6	+79.6

Table 6: Expected seconds saved against the egocentric courier: as predicted on the planner's model of the groups, as predicted on ePlanSys's own schedule of the same policies (`tools/schedule.py`), and as measured.

The perspective courier had the least expected time in acts I and III, and not in act II. The courier chose its reports on the planner's own model of ePlanSys's groups, which runs two robots' actions together wherever they are independent in the domain. ePlanSys's scheduler, ported to tools/schedule.py and run over the same policies, keeps apart two drives that end at one place and holds a branching look's group until its longest member ends. On that schedule the perspective courier saves 12.7 s in act I, 5.1 s in act II, 72.6 s in act III, against 8.2, 33.4, 98.1 s on the planner's model, and the batch measured 16.7, -3.8, 79.6 s. The report that corrects a robot's revision after a surprise pays on the floor; the one that only steers a route at dusk pays where the schedule leaves a queue to avoid, as with three robots at one lobby, and not where it does not, as with two. Without pooling the runs of identical paths the expected times were act I 285.4, 265.0, 263.7; act II 306.4, 301.9, 310.2; act III 384.2, 386.9, 304.6 s, in the same order. The planner's predicted times exceeded the measured by 39.3 s on average (10 %): the robots drive faster than the planner's 0.17 m/s, and a look takes half its declared seven seconds.

10 Sensitivity

These are predictions, on the planner's own model of the groups. Planned again with the listeners' trust from 1 to 6, the perspective policy's predicted expected time did not change in any act: a report too weak to change what the disinfection robot believes still changes where it goes, because ranks weigh its expected time as orders of magnitude, and the linen robot, ranking nothing, is moved by any trust. Charged under other shares of west-loop nights, the perspective policy's predicted expected time was at most the silent courier's in act I for shares 0.05 to 0.4; in act II for shares 0.0 to 0.4; in act III for shares 0.0 to 0.4.

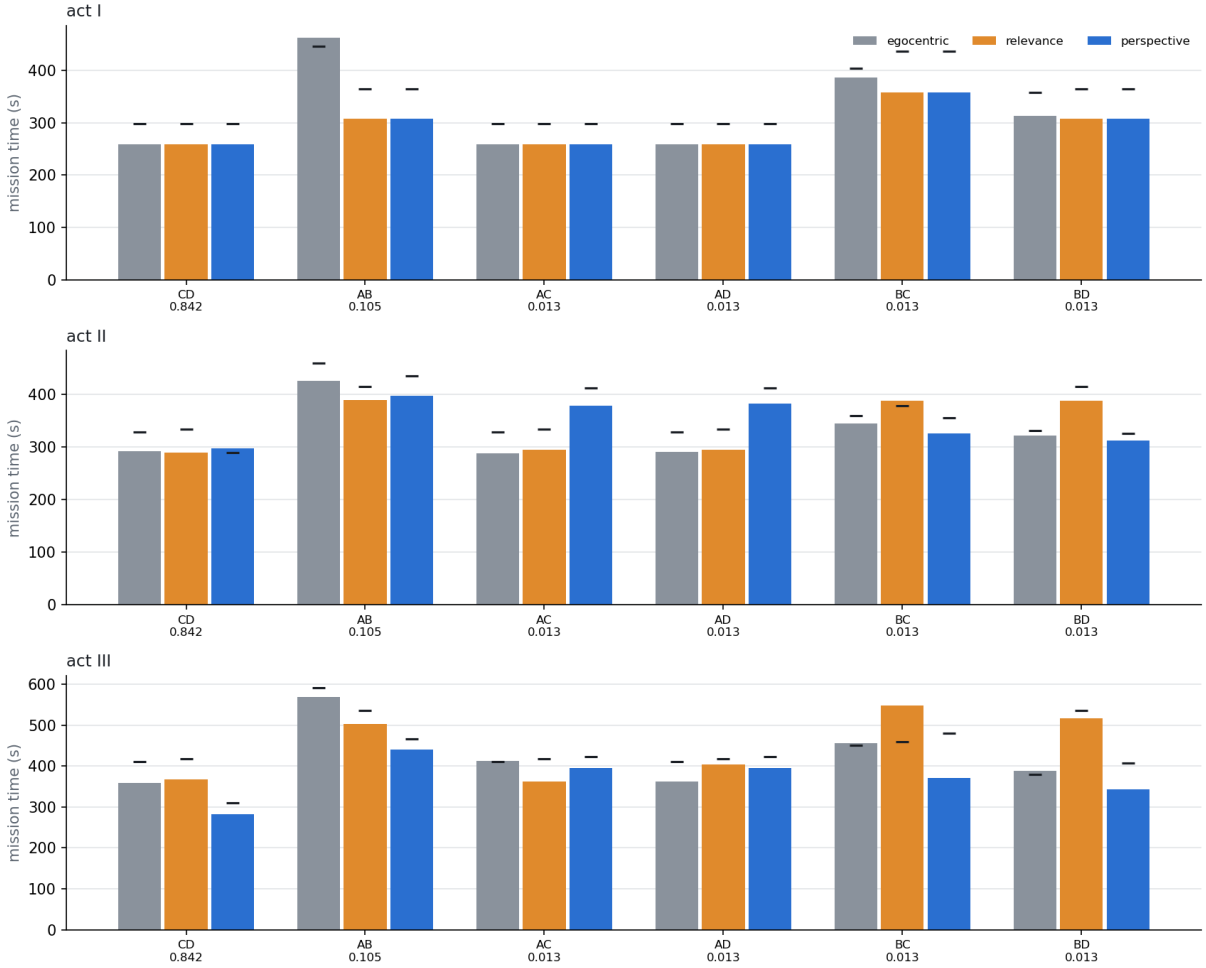


Figure 5: Mission time per night, act and formulation, the mean of the night's runs, pooled with those of formulations that plan the same path; the black tick is the predicted time of the same policy and night. Under each night its frequency.

11 Implementation findings

plank conjoins conditional effects on one atom. The first version of the belief report moved each grade with a conditional effect. plank compiles several conditional effects on one atom into the conjunction of their assignments, so an atom one adds and another deletes loses the addition, and an atom several add needs all their conditions at once. After a report the listener had no grade at the shifted worlds. The policy checker found it at the first report it replayed; the domain now writes one iff per grade atom.

A group is a sequence. The planner first let every robot of a group choose in the same state. An entry chosen so could come, in the group, after another robot's crossing; the crossing told the entering robot that a door was open, which changed what it believed of the door it was entering, and the entry's guard failed where the policy put it. Robots now choose in turn, each in the state the earlier ones' actions left, which is how Aletheia and the executor read the policy.

Looks on another robot's behalf rarely pay on this floor. The perspective courier was first also allowed to look, for the others, at a door it believed shut and a teammate did not, a look with a second-order guard. A lobby visit costs about as much as the trip it saves, and where the rollout chose it, it won mostly by keeping a robot out of a lobby another was queuing for. It was dropped: the courier's only lever is what it says.

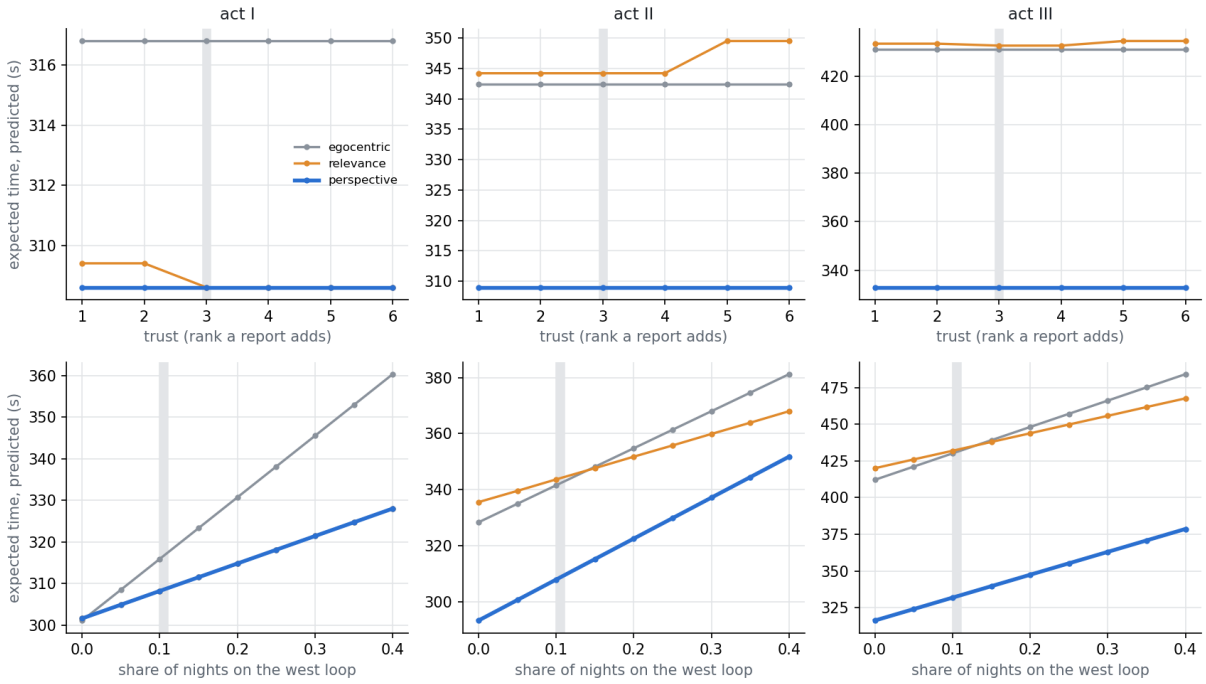


Figure 6: Predicted expected time against the listeners' trust (top) and the share of west-loop nights (bottom), by formulation and act; the grey band marks the stated value.

A robot gave way to a robot standing still. In the first batch, on the west-loop night of act I under the egocentric courier, the courier backing out of lobby B met the disinfection robot, whose drive had ended beside it and which would not move until the courier's drive ended too; the courier gave way, again and again. A robot now gives way only to a robot that is moving, and once in a drive; the batch was run again from the start with that drive.

The executor's groups are not the planner's. ePlanSys dispatches together the actions of a run of the policy that it finds independent, and its test reads every argument of an action as an agent: two robots' drives that end at the same place in the hall never run together, and a branching look holds its group until its longest member, often another robot's delivery, has ended. The planner's model of the groups runs them together, and so overstates what a robot saves by not waiting at a lobby. A test that named each robot's places apart, so that their drives could run together, put the courier and the linen robot at B's mouth at once on the rostered night of act II; they gave way to each other and backed off for 98 s. The batch keeps the shared names.

Relevance without a stopping rule repeats itself. A courier that tells a belief whenever its listener does not believe it told the same belief again and again at low trust, each report a little short. Relevance now says a belief once to each listener.

12 Limitations

- Only the courier reports beliefs, and trust is fixed and known: no robot learns whose word to take, or lowers its trust when a report proves wrong.
- The evidence here is always public by the time the courier speaks, so a belief report carries no news; it moves the listener through its trust alone. A fleet with private sensing would test the other half of Proposition 5.
- The courier looks one step ahead over the policy without reports; the robots route themselves and do not anticipate what they will be told. The team's policy is not optimised jointly.

- Part of the gain in acts II and III is in the lobbies: the linen robot sent to A is not also waiting behind the courier at B. The policy predicts it because the vocabulary holds each lobby as a resource; the batch measures it.
- Aletheia's own search does not find the first act's policy, of depth 26, within 40 minutes (it reached depth 14); the policies are tools/plan.py's, which Aletheia's plugin validates before every run.
- The courier chose its reports on the planner's model of the executor's groups, not on ePlanSys's own schedule, which tools/schedule.py ports and which predicts the floor better; planning on it is left open.
- One floor, three robots, simulation only.

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